

CENG381 Stochastic Processes

Recurrence Relations — Practice 1

Q1 (30 points). Solve the linear homogeneous recurrence relation with distinct roots:

$$\begin{aligned}a_n &= 5 \cdot a_{n-1} - 6 \cdot a_{n-2}, & n &\geq 2 \\a_0 &= 2, & a_1 &= 5\end{aligned}$$

- Write the characteristic equation. Find its two roots r_1 and r_2 .
 - Write the general solution $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$. Apply the initial conditions to find α_1 and α_2 .
 - Write the closed-form solution. Compute a_4 . As $n \rightarrow \infty$, which root dominates? What does this imply about the growth rate of a_n ?
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Q2 (35 points). Solve the linear homogeneous recurrence relation with a repeated root:

$$\begin{aligned}a_n &= 6 \cdot a_{n-1} - 9 \cdot a_{n-2}, & n &\geq 2 \\a_0 &= 1, & a_1 &= 6\end{aligned}$$

- Write the characteristic equation. Show it has a repeated root $r = 3$.
 - For a repeated root r , the general solution is $a_n = (\alpha_1 + \alpha_2 n) \cdot r^n$. Apply the initial conditions to find α_1 and α_2 .
 - Write the closed-form solution. Compute a_3 . How does the growth rate compare to the distinct-root case in Q1?
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Q3 (35 points). Solve the linear non-homogeneous recurrence relation:

$$\begin{aligned}a_n &= 4 \cdot a_{n-1} + 3n, & n &\geq 2 \\a_1 &= 2\end{aligned}$$

- Ignore the non-homogeneous term for now. Solve the homogeneous part $a_n = 4 \cdot a_{n-1}$ to get the homogeneous solution $a_n^{(h)}$.
- Find a particular solution by substituting $a_n^{(p)} = cn + d$ into the original recurrence. Determine the constants c and d .
- The general solution is $a_n = \alpha_1 4^n + cn + d$. Apply the initial condition $a_1 = 2$ to find α_1 . Verify by computing a_2 directly from the recurrence.